

ESG and Greenwashing Detection in Corporate Disclosures Using Natural Language Processing



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Problem Definition

Problem Statement

- Corporate sustainability reports often contain vague, non-measurable ESG claims, making it difficult to evaluate transparency and accountability.
- Manual analysis of ESG disclosures is time-consuming, subjective, and not scalable, especially with long PDF reports and 10-K filings.
- There is no unified automated system that can classify ESG content, analyze sentiment, summarize key topics, and detect potential greenwashing.



Proposed Solution

- Develop a complete NLP pipeline that extracts text, segments sentences, and performs ESG pillar classification, sentiment analysis, and summarization.
- Build a greenwashing detection module that identifies measurable vs. vague ESG statements using transformer-based models.
- Provide interpretable outputs (Excel sheets, summaries, and metrics) to help users quickly understand ESG focus areas and transparency levels.

Background and Prior work



**Growing ESG Reporting
Requirements**



**Greenwashing Concerns in
Literature**



**NLP for Sustainability
Text Analysis**

Gap in Literature



Sentiment Only



**Basic Keyword
Extraction**



**General Purpose
Classification**

Opportunity Identified

**No holistic system combining ESG classification + sentiment +
summarization + greenwashing detection**

DATASET



Sustainability reports &
10-K annual filings



User-uploaded PDFs via
interface



Model outputs saved as
structured Excel files



PREPROCESSING



PDF text extraction &
Sentence segmentation
(spaCy)

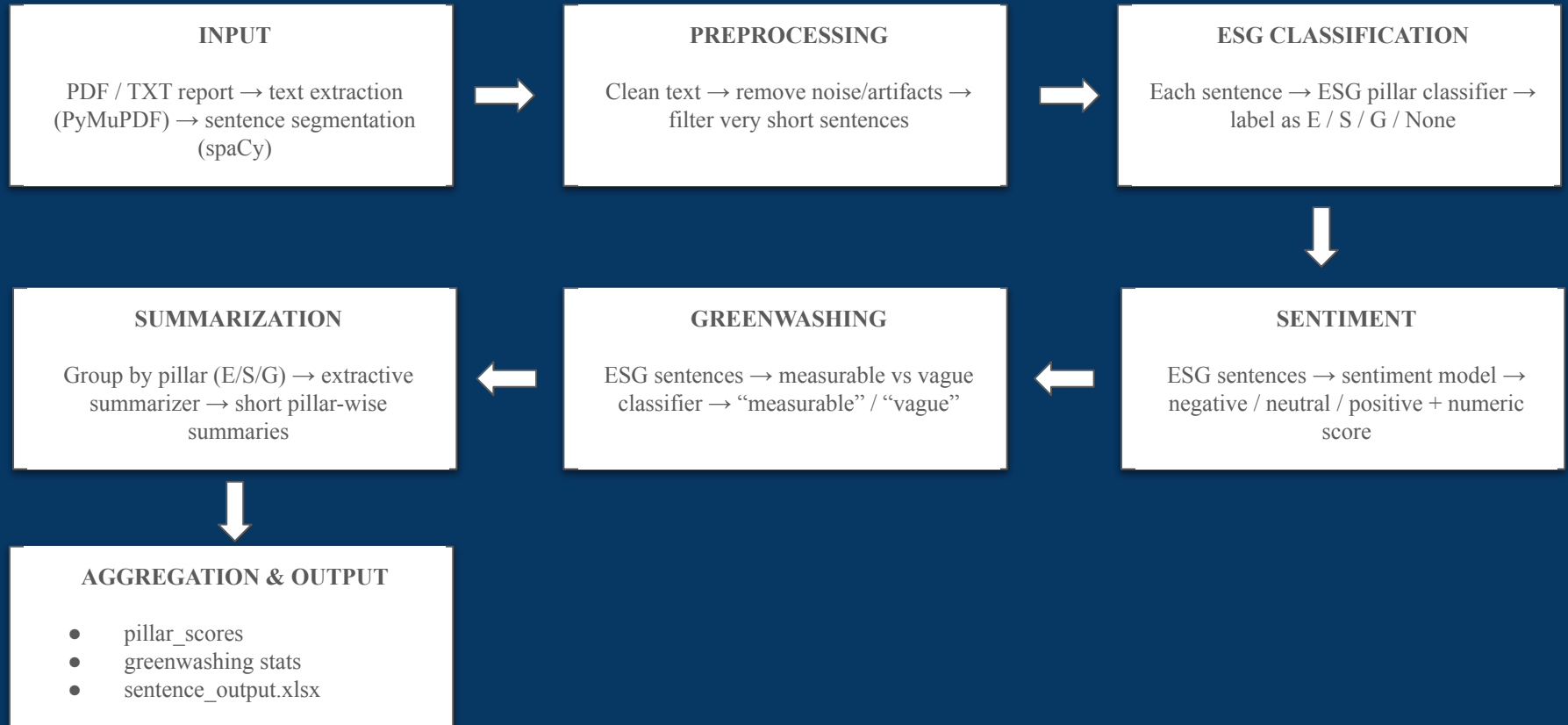


Text cleaning: Unicode
removal, punctuation,
whitespace cleanup



Removing noise: PDF
artifacts, numerical-only
strings

Solution Architecture



Experimental Setup



Environment

- Python (spaCy, PyMuPDF, HuggingFace Transformers)
- CPU runtime (optional GPU for faster inference)
- Batch size: 16-32
- Max token length: 128



Models Used

- ESG Pillar Classifier (Transformer-based)
- Sentiment Classifier (FinBERT-like)
- Greenwashing Detector (Measurable vs Vague)
- Extractive Summarizer (BART/DistilBART)



Evaluation Procedure

- Per-sentence prediction for pillar, sentiment, and measurability
- Pillar-wise aggregation of metrics
- ESG summaries generated independently for E/S/G
- Statistics exported to Excel for analysis

KEY FINDINGS

**ESG sentences are predominantly neutral
(80-90%)**

Pillar distribution:

$E \approx 40-45\%$, $S \approx 25-35\%$, $G \approx 20-30\%$

Greenwashing detection:

- **25-30% of ESG sentences marked as vague**
- **Common vague structures: “We aim to...”, “We strive to...”, “Our commitment to...”**

**Extractive summaries successfully highlight
ESG focus areas**

ERROR ANALYSIS

- **PDF text extraction introduces artifacts (broken words, formatting issues)**
- **Some long sentences truncated at 128 token limit**
- **Summaries sometimes miss domain-specific nuances**
- **Pillar classifier occasionally mislabels multi-topic sentences**

Conclusions



CONCLUSIONS

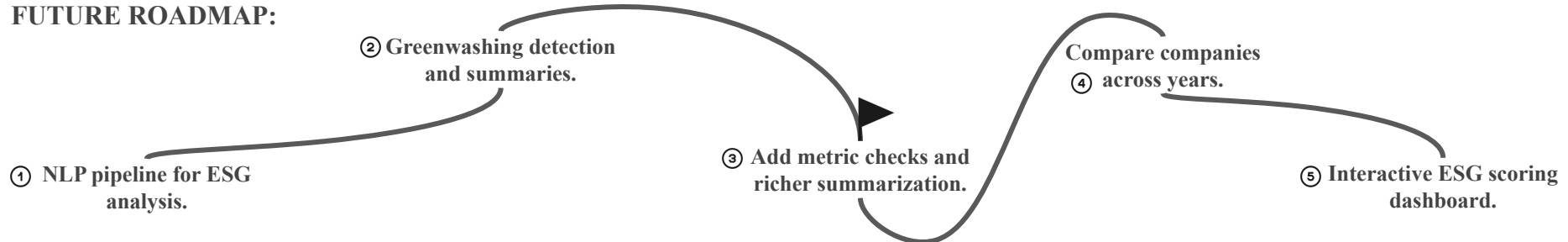
- Built a full NLP pipeline for ESG analytics and greenwashing detection
- System provides actionable insights into ESG transparency
- Helps stakeholders identify vague vs measurable ESG claims



LIMITATIONS

- PDF extraction errors reduce accuracy
- Models may struggle with highly domain-specific corporate jargon
- Does not evaluate numerical consistency across reports

FUTURE ROADMAP:



Thank You